

1. The squeeze theorem

Theorem. Let f, g, h be defined on a common domain D , with points of $D \setminus \{a\}$ arbitrarily close to a . Suppose there is an $r > 0$ such that

$$f(x) \leq g(x) \leq h(x) \quad (x \in D, 0 < |x - a| < r).$$

If

$$\lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} h(x),$$

then $\lim_{x \rightarrow a} g(x) = L$. All three limits use the same domain. The same statement holds for one-sided limits.

Proof. Let $\varepsilon > 0$. The assumed limits give $\delta_f, \delta_h > 0$ such that, for $x \in D$,

$$\begin{aligned} 0 < |x - a| < \delta_f &\implies |f(x) - L| < \varepsilon, \\ 0 < |x - a| < \delta_h &\implies |h(x) - L| < \varepsilon. \end{aligned}$$

Choose $\delta = \min\{r, \delta_f, \delta_h\}$. If $0 < |x - a| < \delta$, the ordering and both estimates apply, giving

$$L - \varepsilon < f(x) \leq g(x) \leq h(x) < L + \varepsilon.$$

Thus $|g(x) - L| < \varepsilon$, as required. $\square$

Example

For $x \neq 0$, the bound $-1 \leq \cos(1/x) \leq 1$ gives

$$-x^2 \leq x^2 \cos(1/x) \leq x^2.$$

Since both bounds tend to zero,

$$\boxed{\lim_{x \rightarrow 0} x^2 \cos(1/x) = 0.}$$

No limit of $\cos(1/x)$ is needed. Define

$$G(x) = \begin{cases} x^2 \cos(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

The limit we just proved equals $G(0)$, so G is continuous at zero.

The same conclusion follows directly from the definition. Given $\varepsilon > 0$, choose $\delta = \sqrt{\varepsilon}$. Then

$$|x| < \delta \implies |G(x) - G(0)| \leq x^2 < \varepsilon.$$

This includes $x = 0$.

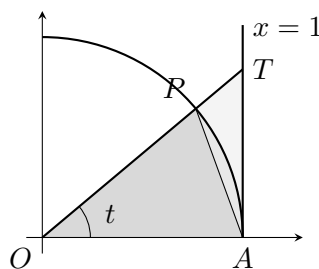
2. Trigonometric limits

All angles in this section are measured in **radians**. We use the usual unit-circle definitions and trigonometric identities.

Geometric inequality. Let $0 < t < \pi/2$. In the unit circle let

$$O = (0, 0), \quad A = (1, 0), \quad P = (\cos t, \sin t).$$

Extend the ray OP to meet the vertical line $x = 1$ at $T = (1, \tan t)$.



The triangle OAP lies inside the sector OAP , which lies inside the triangle OAT . Their areas give

$$\frac{1}{2} \sin t < \frac{1}{2} t < \frac{1}{2} \tan t, \quad \text{so} \quad \boxed{\sin t < t < \tan t.}$$

The sector area is $t/2$: a full angle 2π has area π . This argument uses geometry, not continuity or differentiation.

Limits at zero. The geometric inequality, oddness of sine, and $\sin 0 = 0$ imply

$$|\sin t| \leq |t| \quad (|t| < \pi/2).$$

Squeezing between $-|t|$ and $|t|$ gives $\sin t \rightarrow 0$. For $|t| < \pi/2$, the half-angle identity gives

$$0 \leq 1 - \cos t = 2 \sin^2(t/2) \leq \frac{t^2}{2}.$$

Another application of squeeze gives

$$\boxed{\lim_{t \rightarrow 0} \sin t = 0, \quad \lim_{t \rightarrow 0} \cos t = 1.}$$

These limits establish continuity at zero; continuity was not assumed to obtain them.

2. Trigonometric limits (continued)

The fundamental limit. For $0 < t < \pi/2$, we have $\cos t > 0$. The inequalities $\sin t < t < \tan t$ therefore imply

$$\cos t < \frac{\sin t}{t} < 1.$$

Both bounds tend to 1, so the right-hand limit is 1. Since

$$\frac{\sin(-t)}{-t} = \frac{\sin t}{t},$$

the left-hand limit is also 1. By the one-sided criterion,

$$\boxed{\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1.}$$

The cosine estimate on the preceding page also gives

$$\left| \frac{1 - \cos t}{t} \right| \leq \frac{|t|}{2} \quad (t \neq 0) \quad \implies \quad \boxed{\lim_{t \rightarrow 0} \frac{1 - \cos t}{t} = 0.}$$

Continuity. Fix $a \in \mathbb{R}$. The addition formulas and the limits at zero imply

$$\begin{aligned} \sin(a + h) &= \sin a \cos h + \cos a \sin h \longrightarrow \sin a, \\ \cos(a + h) &= \cos a \cos h - \sin a \sin h \longrightarrow \cos a \quad (h \rightarrow 0). \end{aligned}$$

Thus sine and cosine are continuous at every real number. By continuity of quotients, $\tan x = \sin x / \cos x$ is continuous wherever $\cos x \neq 0$. The same reasoning applies to the reciprocal trigonometric functions on their domains.

Example

For $x \neq 0$,

$$\frac{\sin(3x)}{x} = 3 \frac{\sin(3x)}{3x}.$$

As $x \rightarrow 0$, the nonzero number $3x$ tends to zero. Hence

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = 3.$$

More explicitly, if the defining estimate for $\sin t/t$ holds when $0 < |t| < \eta$, it applies here when $0 < |x| < \eta/3$.

3. Infinite limits

Let f be defined on a deleted interval around a . We write

$$\lim_{x \rightarrow a} f(x) = +\infty$$

if, for every $M > 0$, there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \implies f(x) > M.$$

Similarly, $\lim_{x \rightarrow a} f(x) = -\infty$ means that for every $M > 0$ some $\delta > 0$ makes

$$0 < |x - a| < \delta \implies f(x) < -M.$$

The symbols $+\infty$ and $-\infty$ are not real numbers. These statements describe specific ways in which a finite real limit fails to exist. For one-sided infinite limits replace $0 < |x - a| < \delta$ by $0 < x - a < \delta$ or $0 < a - x < \delta$.

Example

We prove

$$\boxed{\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty.}$$

Given $M > 0$, choose $\delta = 1/\sqrt{M}$. Then

$$0 < |x| < \delta \implies 0 < x^2 < \frac{1}{M} \implies \frac{1}{x^2} > M.$$

This also excludes any finite limit: a function with a finite limit is bounded on some deleted interval, whereas this estimate forces $1/x^2$ above every fixed bound near zero.

Example

For $f(x) = 1/x$, choosing $\delta = 1/M$ gives

$$0 < x < \delta \implies \frac{1}{x} > M, \quad -\delta < x < 0 \implies \frac{1}{x} < -M.$$

Therefore

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

There is no two-sided finite limit, and neither two-sided infinite limit holds. Values on the two sides have opposite signs.

Vertical asymptotes. The line $x = a$ is a vertical asymptote if at least one of the one-sided limits of f at a is $+\infty$ or $-\infty$. Both examples have the vertical asymptote $x = 0$. A denominator equal to zero is not sufficient by itself: $(x^2 - 1)/(x - 1) = x + 1$ for $x \neq 1$, with finite limit 2 at 1.

4. Limits at infinity

Let f be defined for all sufficiently large positive x . The statement $\lim_{x \rightarrow +\infty} f(x) = L$, with $L \in \mathbb{R}$, means

for every $\varepsilon > 0$ there is an $A > 0$ such that $x > A \implies |f(x) - L| < \varepsilon$.

For $\lim_{x \rightarrow -\infty} f(x) = L$, assume f is defined for all sufficiently negative x and replace $x > A$ by $x < -A$. An infinite limit at infinity uses M instead of ε : for example,

$$\lim_{x \rightarrow +\infty} f(x) = +\infty \iff \forall M > 0 \exists A > 0 \forall x > A : f(x) > M.$$

For a limit of $-\infty$, use $f(x) < -M$; for $x \rightarrow -\infty$, use $x < -A$. Thus $x^2 \rightarrow +\infty$ at both ends: take $A = \sqrt{M}$.

Example

For $\varepsilon > 0$, choose $A = 1/\varepsilon$. If $x > A$ or $x < -A$, then

$$\left| \frac{1}{x} \right| = \frac{1}{|x|} < \frac{1}{A} = \varepsilon.$$

Consequently $1/x \rightarrow 0$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. The finite-limit laws apply at infinity too: the same estimates work, with a maximum of lower thresholds A in place of a minimum of neighborhood sizes δ . In particular, $1/x^2 \rightarrow 0$.

Horizontal asymptotes. The line $y = L$ is a horizontal asymptote if $\lim_{x \rightarrow +\infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$. The two ends must be considered separately.

Example

Divide numerator and denominator by x^2 :

$$\frac{3x^2 - x}{x^2 + 1} = \frac{3 - 1/x}{1 + 1/x^2} \longrightarrow \frac{3 - 0}{1 + 0} = 3 \quad (x \rightarrow \pm\infty).$$

Thus $y = 3$ is a horizontal asymptote at both ends. This graph can cross its asymptote: equality with 3 holds at $x = -3$. An asymptote describes a limit, not a barrier to the graph.

Oblique asymptotes. More generally, if $f(x) - (mx + b) \rightarrow 0$ as $x \rightarrow +\infty$ or $-\infty$, then $y = mx + b$ is an asymptote at that end. For example,

$$\frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}$$

has the oblique asymptote $y = x + 1$ at both ends.

5. The derivative

Let f be defined on an open interval containing a . For $h \neq 0$ sufficiently small, the quantity

$$\frac{f(a+h) - f(a)}{h}$$

is the slope of the secant line through $(a, f(a))$ and $(a+h, f(a+h))$. It is also the average rate of change of f over these two inputs.

Definition. The derivative of f at a is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists as a finite real number. In that case f is *differentiable at a* . Equivalently,

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

The number $f'(a)$ is the instantaneous rate of change. The tangent line at $(a, f(a))$, when this finite derivative exists, has equation

$$y = f(a) + f'(a)(x - a).$$

Here the definition uses a two-sided limit at an interior point.

Example

For $f(x) = x^2$ and $h \neq 0$,

$$\frac{f(a+h) - f(a)}{h} = \frac{(a+h)^2 - a^2}{h} = 2a + h \rightarrow 2a.$$

Hence $f'(a) = 2a$ for every $a \in \mathbb{R}$. At $a = 2$, the tangent line is $y = 4 + 4(x - 2) = 4x - 4$. The secant slope is $4 + h$; as h tends to zero through positive or negative values, it tends to the same number 4.

Reading. Stewart, *Calculus: Early Transcendentals*, 9th ed.: §2.3 (squeeze theorem), §3.3 (trigonometric limits), §2.2 and §2.6 (infinite limits, limits at infinity and asymptotes), §2.7 (the derivative at a point).